Green Public Scheduling and Route Optimization

CSA0620- Design and analysis of algorithms

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INTRODUCTION

- Public transportation plays an important role in daily urban travel.
- Poor route planning and scheduling can increase passenger waiting times.
- Unnecessary bus trips increase operating costs and emissions.
- Some rural and underserved communities have limited transport access.
- Algorithm-based optimization can improve route efficiency and service quality.
- The project focuses on sustainable, accessible, and efficient public transportation.

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OBJECTIVES

- Reduce waiting time: Minimize the average time passengers wait for buses.
- Optimize routes: Identify efficient paths between bus stops.
- Reduce vehicle kilometres: Minimize unnecessary travel distances.
- Improve scheduling: Allocate available buses according to passenger demand.
- Ensure accessibility: Maintain adequate service to underserved communities.
- Evaluate performance: Compare the existing and optimized transport plans.

PROBLEM STATEMENT

- Existing transport systems may use inefficient routes and schedules.
- Passengers may experience long waiting times and inconvenient connections.
- Poor bus allocation can lead to underutilized or overcrowded vehicles.
- Excessive vehicle kilometres increase fuel consumption and emissions.
- Underserved communities may receive insufficient transport services.
- A system is needed to optimize routes and schedules while maintaining accessibility.

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EXISTING SYSTEM

- Traditional public-service scheduling and route planning are often performed manually or using fixed schedules.
- Routes may not be updated according to traffic conditions, service demand, or road-network changes.
- Inefficient route planning can increase travel distance, time, and fuel consumption.
- Higher fuel consumption can lead to increased carbon emissions and environmental pollution.
- Manual scheduling may result in poor resource utilization and delays in service delivery.

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PROPOSED SYSTEM

- Collect bus-stop locations, route distances, travel times, and passenger demand.
- Represent the transport network as a graph, where stops are vertices and connections are edges.
- Apply shortest-path algorithms to identify efficient routes.
- Use greedy algorithms to assign buses and select promising departure times.
- Apply dynamic programming to optimize scheduling decisions under defined constraints.
- Calculate waiting time, vehicle kilometres, emissions, and service coverage.
- Compare the optimized plan with the original transport plan.

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PROPOSED ALGORITHM'S

Algorithm	Purpose
Dijkstra's Shortest-Path Algorithm	Finds the shortest or lowest-cost path between bus stops.
Greedy Algorithm	Makes scheduling and vehicle-allocation decisions using a defined priority rule.
Dynamic Programming	Evaluates scheduling alternatives to minimize total cost under constraints.

Module 1: Route Network Management and Shortest-Path Optimization

- Store bus-stop names, locations, and IDs.
- Represent bus stops as vertices and routes as edges in a graph.
- Record distances and travel times between bus stops.
- Use Dijkstra's algorithm to find the shortest route.
- Identify alternative routes when needed.
- Calculate total route distance and estimated travel time.
- Reduce unnecessary travel and improve route efficiency.

Module 2: Intelligent Scheduling and Vehicle Allocation

- Collect passenger demand and bus availability data.
- Allocate buses to routes based on passenger demand.
- Optimize bus departure times and service frequency.
- Use Greedy algorithms to make scheduling decisions.
- Apply Dynamic Programming to optimize scheduling alternatives.
- Reduce passenger waiting time and improve vehicle utilization.
- Avoid overcrowding and unnecessary bus trips.

Module 3: Emission Analysis and Accessibility Monitoring

- Calculate total vehicle kilometres travelled by buses.
- Estimate carbon emissions based on distance and fuel or energy consumption.
- Monitor passenger waiting time and transport efficiency.
- Measure public transport coverage across different communities.
- Identify underserved areas with limited transport access.
- Compare existing and optimized transport plans.
- Reduce emissions while maintaining accessibility for all communities.

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Advantages

- More efficient public transport routes.
- Better use of available buses.
- Improved scheduling based on passenger demand.
- Reduced unnecessary vehicle travel.
- Better transport accessibility for underserved communities.
- Data-driven evaluation of transport performance.

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Conclusion

- The project uses algorithms to improve public transport routing and scheduling.
- Dijkstra's algorithm supports efficient route finding.
- Greedy algorithms support quick scheduling and vehicle-allocation decisions.
- Dynamic programming helps evaluate scheduling alternatives.
- Performance metrics measure efficiency, environmental impact, and accessibility.
- The system aims to balance sustainability with the transport needs of all communities.

THANK YOU